Biological Sciences

	Normative 4-year Degree
Major	Biological Sciences 生物科學
Degree / Award Title	Bachelor of Science in Biological Sciences 理學士(生物科學)
Offering Academic Unit	Department of Biomedical Sciences
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2401 English for Science	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12
College-specified Courses	PHY1400 Introductory Physics for Biologist	3
	CHEM1200 Discovery in Biology	3

	CHEM1300 Principles of General Chemistry	3

College / School Requirements (0 credit units) **Catalogue Term : Semester A 2025/26**

Not required.

Major Requirements (72 credit units) **Catalogue Term : Semester A 2026/27**

Notes:

The catalogue term of the major requirement that students will follow will be the effective term of the declared/allocated major.

For normative 4-year degree students who will join the majors allocation exercise, the catalogue term of major requirement will be one year after admission.

For advanced standing students and 4-year degree students who already have a major at the time of admission, the catalogue term of major requirement will be the same as their admission term.

1. Core Courses (57 credit units)

Course Code	Course Title	Credit Units
BMS1901	Calculus for Life Sciences	3
BMS2004	Biochemistry	3
BMS2005	Human Physiology	3
BMS2202	Diversity of Life and Evolution	3
BMS2204	Diversity of Life and Microbiology Laboratory	2
BMS2205	Essential Techniques in Biomedical Sciences	4
BMS2206	Cell Biology	3
BMS2901	Introductory Biostatistics and Data Analysis	3
BMS3203A	Genetics	3
BMS3204	Molecular Biology	3
BMS3205	Omics, Genome Editing and Advanced Techniques for Biomedical Research	2
BMS3301	Bioinformatics	3
BMS4007	Pharmacology and Toxicology	3
BMS4008	Clinical Immunology	3
BMS4102	Technology for Regenerative Medicine	3
BMS4301	Cancer Biology	3
BMS4303	Neuroscience	3
CHEM2013	Microbiology	3
CHEM3068	General Ecology	4

2. Stream Core Course (15 credit units) (Human and Artificial Intelligence Stream)

On top of the core courses (57 credit units) listed above, students who opt for Human and Artificial Intelligence Stream are required to take the following stream core courses (15 credit units), in total 72 credit units for core courses.

Course Code	Course Title	Credit Units
NS2003	Introduction to Artificial Intelligence with Brain Computing and Application	3
NS3003	Ethical Application of Artificial Intelligence in Biological Sciences and Healthcare	3
NS4002	A Mathematical Introduction to Neural Networks	3
NS4003	Practical Artificial Intelligence Internship in Life Sciences and Biotech	6

3. Electives (15 credit units)

1. Students are required to register either BMS4206 Final Year Project in Biomedical Research **OR** BMS4304 Industrial Attachment: Biotechnology and Health Sciences as one of the Major Elective courses to fulfill university degree requirement.
2. Students who opt for Human and Artificial Intelligence Stream are not required to take any electives.

Course Code	Course Title	Credit Units	Remarks
BMS1701A	Biomedical Research – Rotation Project I (Theme A)	1	
BMS1701B	Biomedical Research – Rotation Project I (Theme B)	1	
BMS1701C	Biomedical Research – Rotation Project I (Theme C)	1	
NS1001	Brain Structure and Function	3	
BMS2002	Pathophysiology	3	
BMS2003B	Clinical Chemistry	2	
BMS2007	Human Anatomy	3	
BMS2008B	Hematology I	2	
BMS2301A	Biomedical Research – Rotation Project II (Theme A)	1	
BMS2301B	Biomedical Research – Rotation Project II (Theme B)	1	
BMS2301C	Biomedical Research – Rotation Project II (Theme C)	1	
NS2001	Cognition and Behavior	3	
NS2003	Introduction to Artificial Intelligence with Brain Computing and Application	3	Will be counted as Stream Core course for those students declared Human and Artificial Intelligence Stream
BMS3002B	Cellular Pathology	2	
BMS3003B	Advanced Clinical Chemistry	2	
BMS3006B	Transfusion Science and Technology	2	
BMS3007	Ethics and Practice in Medical Laboratory	3	
BMS3011B	Hematology II	2	
BMS3101	Cell Transport and Signalling	3	

NS3001	Technologies in Neuroscience	3	
NS3002	Learning and Memory	3	
NS3003	Ethical Application of Artificial Intelligence in Biological Sciences and Healthcare	3	Will be counted as Stream Core course for those students declared Human and Artificial Intelligence Stream
BMS4001	Medical Informatics and Laboratory Management	3	
BMS4003B	Clinical Biochemistry and Molecular Diagnostics	2	
BMS4004B	Advanced Cellular Pathology	2	
BMS4005B	Medical Virology	2	
BMS4206	Final Year Project in Biomedical Research	8	Not to be taken by students who opt for Human and Artificial Intelligence Stream
BMS4304	Industrial Attachment: Biotechnology and Health Sciences	8	Not to be taken by students who opt for Human and Artificial Intelligence Stream
CHEM4078	Aquatic Ecology	4	
NS4001	Brain Disorders and Therapy	3	
NS4002	A Mathematical Introduction to Neural Networks	3	Will be counted as Stream Core course for those students declared Human and Artificial Intelligence Stream
NS4003	Practical Artificial Intelligence Internship in Life Sciences and Biotech	6	Will be counted as Stream Core course for those students declared Human and Artificial Intelligence Stream